

Material Reflectance Property Estimation of Complex Objects Using an Attention Network

Bin Cheng*
School of Artificial
Intelligence,
Beijing Normal University.

Junli Zhao†
College of Computer
Science and Technology,
Qingdao University.

Fuqing Duan‡
School of Artificial
Intelligence,
Beijing Normal University.

ABSTRACT

Material reflectance property modeling can be used in realistic rendering to generate realistic appearances for virtual objects. However, current works mainly focus on near plane objects. In this paper, we propose an end-to-end network framework with attention mechanism to estimate the reflectance properties of any 3D object surface from a single image, where a single attention module is used for each reflectance property respectively to learn the property specific features. We also generate a material dataset by rendering a set of 3D complex shape models. The dataset is suitable for reflectance property estimation of arbitrary complex shape objects. Experiments validate the proposed method.

Index Terms: Computing methodologies—Computer graphics—Reflectance properties estimation; Computing methodologies—Artificial intelligence—Computer Vision—Reconstruction

1 INTRODUCTION

Modeling material reflectance properties is a challenging task. Most of previous methods rely on well-designed and calibrated light field acquisition systems to densely collect the texture photos of the object under different illuminations and viewpoints, and estimate the reflectance properties by inverse rendering of these texture photos. It is commonly a time-consuming and laborious work to collect the data of an object. The material reflectance property of a general object is usually represented by Spatially Varying Bidirectional Reflectance Distribution Function (SVBRDF), which is mainly determined by the object shape and the properties such as diffuse albedo, roughness and specular albedo. As shown in Figure 1, material reflectance property estimation here is to estimate the properties at the 3D object surface point corresponding to each pixel from the collected texture image.

In recent years, reflectance property estimation based on deep learning has gradually gained popularity. Generally, deep learning methods [1, 6] estimate the reflectance properties from sparse multi-view images, and they reduce the complexity of data acquisition and reflectance estimation. However, previous works mainly focus on near plane objects, and the available material datasets that are requisite are very limited.

In this work, we use a set of 3D cultural antique models of complex shapes to generate a material dataset, which can be used for reflectance property estimation of arbitrary complex shape objects, and an end-to-end network framework based on attention mechanism is proposed to estimate the material reflectance properties of any 3D object surfaces from a single image, where the attention mechanism is used to learn property specific features for each reflectance property.

*e-mail: 1401050696@qq.com

†e-mail: zhaojl@yeah.net.com

‡e-mail: fqduan@bnu.edu.cn, corresponding author

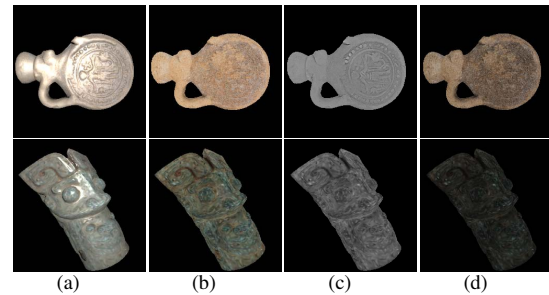


Figure 1: Estimating reflection properties from a single image. (a-d) Input image, diffuse albedo, roughness and specular albedo.

2 DATASET GENERATION

We download 82 cultural relic models with relatively complex shapes or materials from the Sketchfab¹, where some missing reflectance properties are completed manually.

For data augmentation, we perform 22 random linear transformations on the original texture maps of diffuse albedo, roughness and specular albedo respectively and independently. Here we use the microfacet BRDF model proposed by Karis et al [4] and a physically-based method [2] to render the models with the generated reflectance properties in 20 viewpoints, and for each model we respectively use the point light source collocated and the one non-collocated with the camera. In the end, we generated 37720 samples. Each sample is composed of two rendered images, three reflectance property maps of diffuse albedo, roughness, and specular albedo, and two geometric property maps of depth and normal. The resolution of each map is 256*256. A few samples are shown in Figure 2.

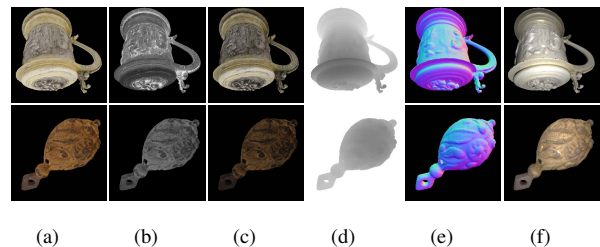


Figure 2: Samples from the generated dataset. (a)-(f) Diffuse albedo, roughness, specular albedo, depth, normals and rendered images.

3 REFLECTANCE PROPERTIES ESTIMATION

The proposed reflectance estimation framework is shown in Figure 3, which includes an initial estimation network(InitNet) and a re-

¹www.cgchannel.com/tag/sketchfab/

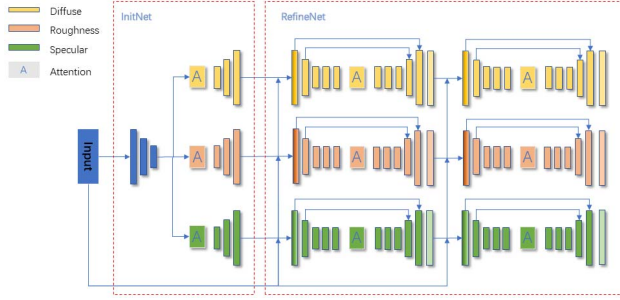


Figure 3: An reflectance properties estimation network. The network is composed of the InitNet and the RefineNet. An attention module is introduced to U-net to learn the feature weights of different properties.

finest estimation network(RefineNet). Unlike the cascade network structure proposed by Li et al. [6], this framework is a multi-task learning framework for estimating the three reflectance properties.

InitNet adopts an auto-encoder structure. The encoded features are shared by three reflectance properties i.e. diffuse reflectance, roughness, and specular reflectance, to utilize the potential correlation among these properties. For the decoding of reflectance properties, a separate attention module is used to learn the property specific attention features for each property.

Following the initial estimation network, a single optimization channel in the refined network is connected to refine the estimation of each reflectance property. The refined network bridges the two network modules through a stack like structure to enhance the learning ability of the network, where each module also adopts a network structure of U-net [5], and the input image is also added to the input of the module to prevent the loss of image information. Like the initial estimation network, an attention module is also added in each auto-encoder to learn the property specific features. In addition, a residual block [3] is added to each auto-encoder to accelerate the training of the network.

Loss Function measures the L2 loss of material reflectance properties and the rendering results of reflectance properties, where each includes the initial estimations and refined estimations. Moreover, in order to reduce the impact of highlights, we also consider the rendering loss under the illumination of the non-collocated point light source.

4 EXPERIMENT

Among the generated 37,720 samples of the 82 models, a total of 30,360 samples of 66 models are used for training, and the remaining 16 models for testing. We first perform the ablation analysis of the refined network, the attention module and the new rendering loss with the non-collocated light source respectively, and then compare our method with Li's work [6]. We use the average L2 errors of the estimations on test data to evaluate the methods. The results are shown in Table 1, We can see that the proposed strategies are effective.

Table 1: Comparisons by L2 errors.

	diffuse	roughness	specular
InitNet	0.068147	0.161329	0.108718
NoAttention	0.056403	0.162340	0.109014
NoNewLoss	0.068409	0.158835	0.102988
RefineNet	0.054880	0.150679	0.106040
Li	0.241817	0.389330	-

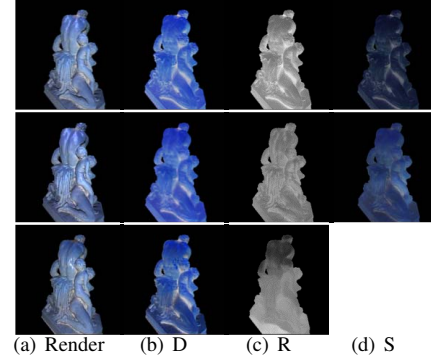


Figure 4: From top to bottom are the ground truth, our results, and Li's results. From left to right are rendered image, diffuse albedo, roughness and specular albedo.

Figure 4 shows an example, where (a), (b), (c) show the ground truth, the estimated results by our method and the results by the method in [6] respectively. We can see that our results are closer to the ground truth. Maybe this is because the method in [6] estimates geometry and SVBRDF simultaneously, which makes the problem more ill-posed. Moreover, our model is an end-to-end network, and the attention mechanism we use can well learn the property specific features, which make our method more effective.

5 CONCLUSION

In this paper, a material dataset is generated by rendering a set of 3D complex shape models, and an end-to-end network framework based an attention mechanism is proposed to estimate the reflectance properties of any 3D object surface from a single image. Compared with existing datasets, the generated dataset is more suitable for material reflectance property estimation of arbitrary complex shape objects. The attention module can learn the property specific features to improve the estimation accuracy. A rendering loss term with a non-collocated light source is introduced to reduce the influence of highlight. The deficiency of this method is that it does not consider the influence of global illumination. The future work will consider the modeling of global illumination.

6 ACKNOWLEDGMENTS

This work was supported by the National Key RD Program of China under Grant No. 2019YFC1521103.

REFERENCES

- [1] S. Bi, Z. Xu, K. Sunkavalli, D. Kriegman, and R. Ramamoorthi. Deep 3d capture: Geometry and reflectance from sparse multi-view images. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [2] B. Burley. Physically-based shading at disney. 2012.
- [3] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. *IEEE*, 2016.
- [4] B. Karis. Real shading in unreal engine 4. Technical report, Epic Games, 2013.
- [5] O. Ronneberger, P. Fischer, and T. Brox. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. Springer, Cham, 2015.
- [6] L. I. Zhengqin, X. U. Zexiang, R. Ramamoorthi, K. Sunkavalli, and M. Chandraker. Learning to reconstruct shape and spatially-varying reflectance from a single image. *ACM Transactions on Graphics*, 37(6):269.1–269.11, 2018.